

Marcel Giovanni Munaba

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Education

Technische Universität München, M.Sc. in Computer Science Apr 2026 – Present

- Specialization in Security and Machine Learning

Technische Universität München, B.Sc. in Computer Science Oct 2022 – Mar 2026

- GPA: 2,3
- **Application field:** Mathematics
- **Elective Courses:** Fundamentals of AI, Introduction to Data Science and Statistical Thinking, Business Analytics and Machine Learning, Applied Regression, Patterns in Software Engineering

Hochschule Kaiserslautern, Internationales Studienkolleg Mar 2022 – Jun 2022

- GPA: 1,5
- Preparatory college for students whose secondary school certificate is not recognized as equivalent to the German Abitur.

Note: In Germany, the range of passing grades is from 1.0 to 4.0, where 1.0 represents the highest mark.

Research

Quantifying Trade-Offs in the Blockchain Trilemma: Empirical Analysis in PoW-based Systems Towards Pareto-Optimal Configurations (Grade: 1,3), February 2026
Bachelor's Thesis

- **Research question:** Identify Pareto-optimal configurations for Proof-of-Work blockchains by measuring the blockchain trilemma subconcepts of decentralization, scalability, and security.
- **Methodology:** Deployed private Bitcoin Core networks with varied configuration parameters, then measured decentralization, scalability, and security.
- **Key Contributions:** Established baseline metric impacts, identified key correlations for Pareto efficiency, and developed a reusable framework for future trilemma analyses.

Experience

Working Student - Information Security, limango GmbH – Munich, Germany November 2025 – Present

- Inspect and analyze IDS alerts to distinguish real threats from false positives.
- Tune and optimize firewall rules based on alert analysis and traffic patterns.

Working Student - IT Helpdesk, limango GmbH – Munich, Germany June 2024 – November 2025

- Delivered desktop support and resolved 10+ technical issues weekly across Windows, macOS, and Linux, earning near-perfect user satisfaction.
- Managed 100+ user accounts, access permissions, security settings, hardware inventory lifecycle, and deployed and maintained company devices using a Mobile Device Management (MDM) tool to ensure compliance and streamline device provisioning.
- Supported onboarding and offboarding for up to 5 employees per month by configuring hardware, software, and access.

iPraktikum - iOS Developer, Interhyp Gruppe – Munich, Germany Oct 2024 – Feb 2025

- Partnered with a team of 8 developers in the *iPraktikum* course at TUM, led by the Chair of Applied Software Engineering in partnership with Apple.
- Engineered *HOMEvision*, a VisionOS app leveraging Google Gemini to extract natural, spoken descriptions of dream property characteristics and automatically translate them into smart search filters.

Projects

BenchBit

- A private Bitcoin testing framework, developed as part of the Bachelor's Thesis titled: *Quantifying Trade-Offs in the Blockchain Trilemma: Empirical Analysis in PoW-based Systems Towards Pareto-Optimal Configurations*.
- Tools Used: Python, Ansible

CensorBot

GitHub

- A simple profanity checker. Implemented a Naive Bayes classifier using scikit-learn and fine-tuned dataset generation with LLM assistance.
- Tools Used: Python

TaxiRL

GitHub

- Developed and compared multiple reinforcement learning agents in OpenAI's Gymnasium environment.
- Tools Used: Python

Note: More projects are available at my [personal website](#).

Technologies

Languages: Java, C, Python, Dart, R, Swift, HTML/CSS, JavaScript

Technologies: SQL, Node.js, Godot, Flutter, Ansible

Organizations

KMKI Bayern (Indonesian Catholic student organization in Bayern) - Food & Beverage Committee, Munich, Bavaria

Aug 2023 – Jul 2024

- Managed and organized teams of cooks and kitchen staff across events with 50+ attendees.

SMAK 1 Cup Committee - Band Competition Division, Jakarta, Indonesia

Oct 2020 – Dec 2020

- Assisted in finding, organizing and providing information to participants.

Languages

Indonesian (Native), English (C1), German (C1)